

XIAOYU HUANG

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Education

Georgia Institute of Technology | Atlanta, GA

B.S. in Computer Engineering, Minor in Aerospace Engineering, Robotics

GPA 4.00/4.00, Faculty Honors for 6 semesters

Relevant Courses: Deep Learning, Probabilistic Graphical Models, Computer Vision,

Deep RL for Intelligent Control, Intro to Optimal Control, Linear Programming

August 2019 – Present

Expected Graduation, May 2023

Selected Awards

- **President's Undergraduate Research Award** 2021
 - Merit based award for undergraduate research funding

Publications

- "Creating a Dynamic Quadrupedal Robotic Goalkeeper with Reinforcement Learning"
Xiaoyu Huang*, Zhongyu Li*, Koushil Sreenath et. al., *IEEE Robotics and Automation Letters*. under review. [\[Paper\]](#) [\[Video\]](#)
- "DNNFuser: Generative Pre-Trained Transformer as a Generalized Mapper for Layer Fusion in DNN Accelerators"
Sheng-Chun Kao*, **Xiaoyu Huang***, Tushar Krishna. *Preprint*, 2022. [\[Paper\]](#)

Research Experience

Skill Transformer

Advisor: Prof. Dhruv Batra

Leading Research Assistant

January 2022 – current

- **Overview:** Leveraging offline RL to train a monolithic, end-to-end policy to solve multi-skill long-horizon tasks, like Habitat Rearrangement, involving high-dimensional observation (vision & sensory) and action (10DoF) spaces, where online algorithms were unable to solve.
- **Methodology:** *Behavior Cloning* with Transformer architecture and trained on datasets containing individual skills (no task-planning necessary). Proposed *skill-embedding* to solve transitions between skills. Introduced *PPO* online finetuning.
- **Performance:** Achieved SoTA performance among monolithic policies in Habitat Rearrangement Challenge 2022. More results in progress.

Dynamic Quadrupedal Robotic Goalkeeper

Advisor: Prof. Koushil Sreenath

Leading Research Assistant

Summer 2022

- **Overview:** Combining agile locomotion with end-effector trajectory tracking to develop soccer goalkeeping skills on a quadrupedal robot. Goalkeeping is challenging as it combines highly dynamic locomotion with precise and fast non-prehensile object manipulation. The project evolved into my first-author submission to RA-L.
- **Methodology:** Utilized *Hierarchical Deep RL framework* with multiple control policies and a planning policy. Each control policy enables the robot's front legs as end-effectors to track a 3D Bezier curve, and the planner determines a desired skill and end-effector trajectory to intercept a ball flying to different regions of the goal. Introduced wide range of domain and delay randomizations for zero-shot sim-to-real transfer.
- **Performance:** Achieved 87.5% success rate in intercepting soccer shots in real experiments, with ball's speeds up to 8m/s at a 1.5m x 0.9m goal. Penalty kicks from human players and a quadrupedal robot shooter are also demonstrated.

RL Layer-Fusion Dataflow

Advisor: Prof. Tushar Krishna

Leading Research Assistant

August – December 2021

- **Overview:** Leveraging RL to enable online scheduling of intra-batch caching over multiple DNN layers, the Layer-Fusion problem, across different accelerators to achieve less latency and higher throughput with the same hardware on CNNs.
- **Methodology:** Utilized *Genetic Algorithm* as the data generator and *Decision Transformer* as the offline RL algorithm.
- **Performance:** Achieved SOTA performance, providing a speedup up to two times the original throughput for an arbitrary accelerator configuration and network structure.

Teaching Experience

ECE 2031 Digital Design Laboratory | **Undergraduate Teaching Assistant**

Spring 2021

COE 2001 Statics | **Grader**

Fall 2020

Selected Projects

Tencent

IEG AI Research Lab

Research Intern

May – August 2021

- **Overview:** Developing a multi-agent game AI in a 3D First Person Shooting game that imposes distinct roles and strategies on the agents.
- **Methodology:** Utilized *PPO* with mutual information sharing in a two-agent competition setting. Proposed self-play to achieve self-supervised learning. Distributed training to hundreds of workers and accelerated more than 40 times.
- **Performance:** Achieved human-player level performance on simple settings, measured by Elo score (common in e-sport).

Active Safety for Autonomous & Semi-Auto Vehicles

Advisor: Prof. Panagiotis Tsiotras

VIP (Vertically Integrated Project)

January – August 2021

- **Methodology & Performance:** Proposed on LSTM network integrated with the physics model to predict model racing car dynamics in turning, bringing up to 5% improvement to the prediction model.

Leadership

GT Design, Build, Fly

Advisor: Mr. Carl Johnson

Aerodynamics Lead

August 2019 – May 2021

- Led aerodynamics design team on designing a wooden RC aircraft with a wingspan of 3m and a max speed of 35m/s.
- Won national second place in AIAA (Aeronautics and Astronautics) DBF competition.

Press Coverage

- “Goalkeeping Robot Dog Tends Its Net Like a Pro”, IEEE Spectrum, Oct 16, 2022 [\[Link\]](#)
- “Watch two Mini Cheetah robots square off on the soccer field”, Tech Crunch, Oct 17, 2022 [\[Link\]](#)
- “A reinforcement learning-based for-legged robotic goalkeeper”, Tech Xplore, Oct 25, 2022 [\[Link\]](#)

Summary of Research Skills

Programming: C++, Golang, Python, MATLAB, TensorFlow, Pytorch, Optimization Toolbox (Casadi)

Platforms: ROS, Linux, Kubernetes, AWS (Amazon Web Services), Embedded System (ARM Cortex)

Languages: Chinese (native), English (fluent)